# Formula Collection

*rendered with Typst*

## 1. Quadratic Formula

*Classic.*

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

## 2. Pythagorean Theorem

*Classic.*

$$c = \sqrt{a^2 + b^2}$$

## 3. Sum of first $n$ Squares

*Classic.*

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

## 4. Law of Cosines

*Classic.*

$$c^2 = a^2 + b^2 - 2ab \cos \angle C$$

## 5. Legendre's formula

*Floors.*

$$\nu_p(n!) = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor$$

## 6. Euler's Identity

*The most beautiful equation in mathematics.*

$$e^{\pi i} + 1 = 0$$

## 7. Euler's Lesser-Known Identity

*Troll.*

$$\lceil e \rceil - \lfloor \pi \rfloor = 0$$

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## 8. Normal Distribution

*Thanks to Martin for correcting this!*

$$\Phi(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

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## 9. Fourier Transform

*17 Equations That Changed the World.*

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \omega} \, \mathrm{d}x$$

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## 10. Wave Equation

*17 Equations That Changed the World.*

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

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## 11. Navier-Stokes Equation

*17 Equations That Changed the World.*

$$\rho\left(\frac{\partial \ \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \ \mathbf{v}\right) = -\nabla p + \nabla \cdot \mathbf{T} + \mathbf{f}$$

---

## 12. Schrodinger's Equation

*17 Equations That Changed the World.*

$$i\hbar \frac{\partial}{\partial t} \Psi = H\Psi$$

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## 13. Black-Scholes Equation

*17 Equations That Changed the World.*

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

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## 14. Relativity

*17 Equations That Changed the World.*

$$E = mc^2$$

---

## 15. Chaos Theory

*17 Equations That Changed the World.*

$$x_{t+1} = kx_t(1 - x_t)$$

---

## 16. Definition of the Derivative

*17 Equations That Changed the World.*

$$\frac{\mathrm{d}\ f}{\mathrm{d}\ x} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

---

## 17. Euler's Formula for Polyhedra

*17 Equations That Changed the World.*

$$V - E + F = 2$$

---

## 18. Gravitation

*17 Equations That Changed the World.*

$$F = \frac{Gm_1m_2}{d^2}$$

---

## 19. AM-GM

*Fun*

$$\frac{x_1 + x_2 + \ldots + x_n}{n} \geq \sqrt[n]{x_1 \cdot x_2 \ldots x_n}$$

---

## 20. Stirling's Approximation

*Fun*

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$$

---

## 21. Stokes' Theorem

*Fun*

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathrm{d}\,\mathbf{S} = \oint_\Gamma \mathbf{F} \cdot \mathrm{d}\,\boldsymbol{\Gamma}$$

---

## 22. Divergence Theorem

*Fun*

$$\iiint_V (\nabla \cdot \mathbf{F})\ \mathrm{d}V = \oiint_S (\mathbf{F} \cdot \mathbf{n})\ \mathrm{d}S$$

---

## 23. Cauchy-Schwarz Inequality

*Fun*

$$|\langle \mathbf{u}, \mathbf{v} \rangle|^2 \leq \langle \mathbf{u}, \mathbf{u} \rangle \cdot \langle \mathbf{v}, \mathbf{v} \rangle$$

---

## 24. Area of a Circle

*Simple*

$$A = \pi r^2$$

---

## 25. Definition of $\tau$

*Troll.*

$$\tau = 2\pi$$

---

## 26. Sophie Germain Identity

*Simple.*

$$a^4 + 4b^4 = (a^2 + 2ab + 2b^2)(a^2 - 2ab + 2b^2)$$

---

## 27. Pascal's Identity

*Classic.*

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

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## 28. Hockey-stick Identity

*Classic.*

$$\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1}$$

---

## 29. Vandermonde's Identity

*Classic.*

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{k}\binom{n}{r-k}$$

---

## 30. Combinations

*Simple.*

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

---

## 31. Heine's Identity

*No idea what this is. Looks cool tho.*

$$\frac{1}{\sqrt{z-\cos\psi}} = \frac{\sqrt{2}}{\pi} \sum_{m=-\infty}^{\infty} Q_{m-\frac{1}{2}}(z) e^{im\psi}$$

---

## 32. Binomial identity

*Classic.*

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

---

## 33. Hermite's Identity

*Hadn't heard of this either.*

$$\sum_{k=0}^{n-1} \left\lfloor x + \frac{k}{n} \right\rfloor = \lfloor nx \rfloor$$

## 34. Matrix Determinant Lemma

*Or this lmao.*

$$\det(\mathbf{A} + \mathbf{u}\ \mathbf{v}^\top) = (1 + \mathbf{v}^\top \mathbf{A}^{-1}\ \mathbf{u}) \det(\mathbf{A})$$

## 35. Euler Product of the Riemann-Zeta function

*Classic.*

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \in \mathbb{P}} \frac{1}{1 - p^{-s}}$$

## 36. Irrationality of the Square Root of 2

*I just really wanted to keep using bb.*

$$\sqrt{2} \notin \mathbb{Q}$$

## 37. Heron's Formula

*Classic.*

$$[\triangle\, ABC] = \sqrt{s(s-a)(s-b)(s-c)}$$

## 38. Heisenberg's Uncertainty Principle

*Classic.*

$$\Delta x \Delta p \approx \hbar$$

## 39. Continued Fraction for $\frac{\pi}{2}$

*@InertialObservr*

$$\frac{\pi}{2} = 1 + \frac{1}{1 + \frac{1}{\frac{1}{2} + \frac{1}{\frac{1}{3} + \frac{1}{\frac{1}{4} + \ddots}}}}$$

## 40. Sophomore's Dream

*Cool.*

$$\int_0^1 x^{-x}\ \mathrm{d}x = \sum_{n=1}^{\infty} n^{-n}$$

---

## 41. Identity involving $\pi$ and $e$

*@InertialObservr*

$$\prod_{n=2}^{\infty} e\left(1 - \frac{1}{n^2}\right)^{n^2} = \frac{\pi}{e\sqrt{e}}$$

---

## 42. Representation of the Golden Ratio

*Classic*

$$\varphi = \sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + \ldots}}}}$$

---

## 43. The Sum of all Positive Integers

*Troll.*

$$\sum_{n=1}^{\infty} n = -\frac{1}{12}$$

---

## 44. Inverse of a complex number

*Gotta know macron man*

$$z^{-1} = \frac{\bar{z}}{|z|^2}, \forall z \neq 0$$

---

## 45. Definition of Convolution

*Shout out to 6.003*

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t - \tau)\ \mathrm{d}\tau$$

---

## 46. Definition of the Kronecker Delta function

*{cases} ftw*

$$\delta_{i,j} = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

---

## 47. Bayes' Theorem

*bae's theorem*

$$P(A \mid B) = \frac{P(B|A)P(A)}{P(B)}$$

---

## 48. Probability Density Function of the Student's $t$-distribution

*fun*

$$f(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

---

## 49. De Morgan's laws

*fun*

$$\neg(P \land Q) \vdash (\neg P) \lor (\neg Q)$$

---

## 50. Principle of Inclusion-Exclusion

*for dummies*

$$|A \cup B| = |A| + |B| - |A \cap B|$$

---

## 51. General Principle of Inclusion-Exclusion

*for galaxy brains*

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{\emptyset \neq J \subseteq \{1,\ldots,n\}} (-1)^{|J|+1} \left|\bigcap_{j \in J} A_j\right|$$

---

## 52. Determinant of a $2 \times 2$ matrix

*{matrix}*

$$\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

---

## 53. Sawtooth Function

*mathbb cases floors, this has it all*

$$S(x) = \begin{cases} x - \lfloor x \rfloor - \frac{1}{2} & x \in \mathbb{R} \setminus \mathbb{Z} \\ 0 & x \in \mathbb{Z} \end{cases}$$

---

## 54. Definition of Graham's Number

*G = g_{64}*

$$g_n = \begin{cases} 3 \uparrow\uparrow\uparrow\uparrow 3 & n = 1 \\ 3 \overset{g_{n-1}}{\uparrow} 3 & n \geq 2 \land n \in \mathbb{N} \end{cases}$$

---

## 55. Burnside's Lemma

*The Lemma that is not Burnside's*

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

---

## 56. Continuum Hypothesis

*independent of ZFC!*

$$\aleph_0 = |\mathbb{N}|, \mathfrak{c} = |\mathbb{R}|$$
$$\nexists A : \aleph_0 < |A| < \mathfrak{c}$$

---

## 57. Spectral Decomposition

*derived from memory*

$$A = \begin{pmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix} \begin{pmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_n \\ | & | & & | \end{pmatrix}^{-1}$$

---

## 58. Pythagorean Identity

*basically just the Pythagorean theorem*

$$\sin^2\theta + \cos^2\theta = 1$$

---

## 59. Double Angle for sin

*back to basics*

$$\sin(2\theta) = 2\sin(\theta)\cos(\theta)$$

---

## 60. Double Angle for cos

*back to basics*

$$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$$

---

## 61. Fermat's Last Theorem

*have a marvelous proof, but this description's too small to contain it*

$$\nexists\{x, y, z, n\} \in \mathbb{N}, n > 2 : x^n + y^n = z^n$$

---

## 62. Fermat's Little Theorem

*fermat's itty bitty theorem*

$$a^p \equiv a(\operatorname{mod} p)$$

---

## 63. Euler's Theorem

*totients*

$$\gcd(a, n) = 1 \Rightarrow a^{\varphi(n)} \equiv 1(\operatorname{mod} n)$$

---

## 64. QM-AM-GM-HM Inequality over 3 variables

*cool-looking*

$$\sqrt{\frac{a^2+b^2+c^2}{3}} \geq \frac{a+b+c}{3} \geq \sqrt[3]{abc} \geq \frac{3}{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}}$$

---

## 65. Extended Law of Sines

*threw in the circumradius as well*

$$\frac{a}{\sin \angle A} = \frac{b}{\sin \angle B} = \frac{c}{\sin \angle C} = 2R$$

---

## 66. Integration by Parts

*it's just the product rule really*

$$\int u \ \mathrm{d}v = uv - \int v \ \mathrm{d}u$$

---

## 67. Definition of Perfect Numbers

*shrug*

$$\left\{ n : \sum_{d \mid n}^{d<n} d = n \right\}$$

---

## 68. Gaussian Integral

*classic trick*

$$\int_{-\infty}^{\infty} e^{-x^2} \ \mathrm{d}x = \sqrt{\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-x^2-y^2} \ \mathrm{d}x \ \mathrm{d}y} = \sqrt{\int_0^{2\pi} \int_0^{\infty} e^{-r^2} r \ \mathrm{d}r \ \mathrm{d}\theta} = \sqrt{\pi}$$

---

## 69. Definition of an Integral

*why not*

$$\int_a^b f(x) \ \mathrm{d}x = \lim_{k \to \infty} \left( (b-a) \sum_{i=1}^{k} \frac{f\left(a + \frac{i(b-a)}{k}\right)}{k} \right)$$

---

## 70. Quantum Fourier transform

*bra ket notation is fun*

$$|x\rangle \mapsto \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_x^k \ |k\rangle$$

---

## 71. Recursive definition of the Hadamard transform

*matrix in cases*

$$H_m = \begin{cases} 1 & m = 0 \\ \frac{1}{\sqrt{2}} \begin{pmatrix} H_{m-1} & H_{m-1} \\ H_{m-1} & -H_{m-1} \end{pmatrix} & m > 0 \end{cases}$$

---

## 72. Wigner Transform of the Density Matrix

*I know some of these words*

$$W(x,p) = \frac{1}{\pi\hbar} \int_{-\infty}^{\infty} \langle x + y \mid \hat{\rho} \mid x - y \rangle e^{-2ipy/\hbar} \, \mathrm{d}y$$

---

## 73. Imaginary numbers

*Just gonna add some simple formulas*

$$i^2 = -1$$

---

## 74. Sum of Cubes

*Simple*

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

---

## 75. RSA Decryption Algorithm

*good ol' rivest*

$$m = c^{e^{-1} \bmod \phi(n)} \pmod{n}$$

---

## 76. Contraposition

*logic yo*

$$(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$$

---

## 77. Equation of a spring

*Gonna use dots like the physicists do*

$$m\ddot{x} = -kx$$

---

## 78. Sum of reciprocals of partial sums of $\mathbb{N}$

*Credit to @IntertialObservr*

$$\sum_{i=2}^{\infty} \frac{1}{\sum_{j=1}^{i} j} = 1$$

---

## 79. Binet's Formula

*Classic*

$$F_n = \frac{1}{\sqrt{5}} \left( \varphi^n - \frac{(-1)^n}{\varphi^n} \right)$$

---

## 80. Sum of first $n$ Cubes

*Classic*

$$\sum_{k=0}^{n} k^3 = \left( \sum_{k=0}^{n} k \right)^2$$

---

## 81. The Basel Problem

*Classic*

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

---

## 82. Root Mean Square

*how could i forget*

$$f_{\text{rms}} = \sqrt{\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} [f(t)]^2 \, \mathrm{d}t}$$

---

## 83. The Harmonic Series

*Classic*

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

---

## 84. Tupper's Self-Referential Formula

*Troll*

$$\frac{1}{2} < \left\lfloor \operatorname{mod}\left( \left\lfloor \frac{y}{17} \right\rfloor 2^{-17\lfloor x \rfloor - \operatorname{mod}(\lfloor y \rfloor, 17)}, 2 \right) \right\rfloor$$

---

## 85. Hölder's Inequality

*Styled like the OTIS handouts by Evan Chen*

$$\left( \sum_{i=1}^{n} a_i \right)^p \left( \sum_{i=1}^{n} b_i \right)^q \geq \left( \sum_{i=1}^{n} \sqrt[p+q]{a_i^p b_i^q} \right)^{p+q}$$

---

## 86. Rearrangement Inequality

*kinda cool*

$$a_1 \leq a_2 \leq \ldots \leq a_n, b_1 \leq b_2 \leq \ldots \leq b_n \Rightarrow \sum_{i=1}^{n} a_i b_i \geq \sum_{i=1}^{n} a_{\sigma(i)} b_i \geq \sum_{i=1}^{n} a_{n+1-i} b_i$$

---

## 87. Power Mean

*like RMS-AM-GM-HM but like generalized*

$$M_{r(x_1, x_2, \ldots, x_n)} = \begin{cases} \left( \frac{1}{n} \sum_{i=1}^{n} x_i^r \right)^{\frac{1}{r}} & r \neq 0 \\ \sqrt[n]{\prod_{i=1}^{n} x_i} & r = 0 \end{cases}$$

---

## 88. Law of Tangents

*yes this actually exists*

$$\frac{a-b}{a+b} = \frac{\tan\left( \frac{\angle A - \angle B}{2} \right)}{\tan\left( \frac{\angle A + \angle B}{2} \right)}$$

---

## 89. Euler's Arctangent Identity

*dammit euler OP*

$$\tan^{-1}\left( \frac{1}{x} \right) = \tan^{-1}\left( \frac{1}{x+y} \right) + \tan^{-1}\left( \frac{y}{x^2 + xy + 1} \right)$$

---

## 90. The Dirichlet Convolution

*bruh*

$$(f * g)(n) = \sum_{d \mid n} f(d) g\left(\frac{n}{d}\right)$$

---

## 91. Sum of a Row of Pascal's Triangle

*not sure how else to word it*

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \ldots + \binom{n}{n} = 2^n$$

---

## 92. Alternating Harmonic Series

*First use of ln*

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \ldots = \ln 2$$

---

## 93. Definitions of Catalan's Constant

*Credit to /u/heropup*

$$G = \beta(2) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2} = \iint_{[0,1]^2} \frac{\mathrm{d}x \ \mathrm{d}y}{1 + x^2 y^2}$$

---

## 94. Series Representation of Apéry's Constant

*Credit to /u/heropup*

$$\zeta(3) = \frac{5}{2} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3 \binom{2n}{n}}$$

---

## 95. Definition of the Euler-Mascheroni Constant

*Credit to /u/heropup*

$$\gamma = \lim_{n \to \infty} \left( \sum_{k=1}^{n} \frac{1}{k} - \ln n \right) = \int_1^{\infty} \left( \frac{1}{\lfloor x \rfloor} - \frac{1}{x} \right) \mathrm{d}x$$

---

## 96. Mertens' therorem

*actually his third theorem*

$$\prod_{p\in\mathbb{P}}^{n}\left(1-\frac{1}{p}\right)\sim\frac{e^{-\gamma}}{\log}n$$

---

## 97. Green's First Identity

*Credit to Varge*

$$\int_{\Omega}(\psi\Delta\varphi+\nabla\psi\cdot\nabla\varphi)\,\mathrm{d}V=\oint_{\partial\Omega}\psi(\nabla\varphi\cdot\mathbf{n})\,\mathrm{d}S$$

---

## 98. Cauchy-Riemann Equations

*complex analysis is best analysis (1); credit to blu_bird*

$$\frac{\partial u}{\partial x}=\frac{\partial v}{\partial y},\frac{\partial u}{\partial y}=-\frac{\partial v}{\partial x}$$

---

## 99. Cauchy's Integral Formula

*complex analysis is best analysis (2); credit to blu_bird*

$$f(z_0)=\frac{1}{2\pi i}\oint_{\Gamma}\frac{f(z)}{z-z_0}\,\mathrm{d}z$$

---

## 100. Cauchy's Differentiation Formula

*complex analysis is best analysis (3); credit to blu_bird*

$$f^{(k)}(z_0)=\frac{k!}{2\pi i}\oint_{\Gamma}\frac{f(z)}{(z-z_0)^{k+1}}\,\mathrm{d}z$$

---

## 101. Functional Equation for the Riemann-Zeta Function

*This is the simplest example of a functional equation in the Langlands program. Conjecturally all Hasse-Weil zeta functions have Euler factorizations and functional equations with the Riemann zeta function as just one example.*

$$\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s)=\pi^{-\frac{1-s}{2}}\Gamma\left(\frac{1-s}{2}\right)\zeta(1-s)$$

---

## 102. Well-ordering Principle

*Classic. Credit to Eucrue*

$$\forall M(M \subset \mathbb{N} \land M \neq \emptyset \Rightarrow \exists m_0[m_0 \in M \land \forall n(n \in M \Rightarrow m \leq n)])$$

---

## 103. Asymptotic Formula for the Dirichlet Divisor Function

*very cool dirichlet*

$$\sum_{n \leq x} \tau(n) = x \log x + (2\gamma - 1)x + O\left(\sqrt{x}\right)$$

---

## 104. Prime Number Theorem

*trivial*

$$\pi(x) \sim \frac{x}{\log} x$$

---

## 105. Cumulative Distribution Function of the Gaussian Distribution

*dense*

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{t^2}{2}} \, \mathrm{d}t$$

---

## 106. Chernoff Bound

*I never really learned what this was*

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[e^{\lambda X}]}{e^{\lambda t}}$$

---

## 107. Union Bound

*Never learned what this was either*

$$\mathbb{P}\left(\bigcup_{i=1}^{n} X_i\right) \leq \sum_{i=1}^{n} \mathbb{P}(X_i)$$

---

## 108. Law of Total Probability

*shrug*

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$$

---

## 109. Linear Least Squares Estimator

*i love regression analysis*

$$L[X|Y] = \mathbb{E}[X] + \frac{\operatorname{cov}(X,Y)}{\operatorname{var}(Y)}(Y - \mathbb{E}[Y])$$

---

## 110. Rademacher Complexity

*The empirical Rademacher complexity of a function class*

$$\mathcal{R}_n(\mathcal{F}) = \mathbb{E}_\varepsilon \left[ \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i f(x_i) \right]$$

---

## 111. Definition of the Dilogarithm

*aka Spence's function. don't wanna be accused of sleeping on spence*

$$\operatorname{Li}_2(z) = -\int_0^z \frac{\log(1-t)}{t}\,\mathrm{d}t,\, z \in \mathbb{C}$$

---

## 112. Leibniz's Determinant Formula

*Determinant of an n by n matrix*

$$\det(A) = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^{n} A_{i,\sigma(i)}$$

---

## 113. Euler-Lagrange Equations

*The basis for all of Lagrangian mechanics*

$$\frac{\partial L}{\partial q_i} = \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{q}_i}$$

---

## 114. Definition of the Euler Totient Function

*what does totient mean anyways?*

$$\varphi(n) = \left|\{k \in \mathbb{N}_{\leq n} \mid \gcd(k,n) = 1\}\right| = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

---

## 115. Sum of Divisors

*i guess this person likes multiplicative functions*

$$\sigma(n) = \sum_{d|n} d = \prod_{p^a \,\|\, n} \frac{p^{a+1} - 1}{p - 1}$$

---

## 116. Einstein Field Equations

*This form makes use of the Einstein tensor*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

---

## 117. Second Fundamental Theorem of Calculus

*credit to VBG*

$$\int_a^b f(x)\ \mathrm{d}x = [F(x)]_a^b = F(b) - F(a)$$

---

## 118. Abel's Summation Formula

*unclear to me why this is at all useful tbh*

$$\sum_{x<n\leq y} a(n)f(n) = A(y)f(y) - A(x)f(x) - \int_x^y A(t)f'(t)\ \mathrm{d}t$$

---

## 119. Lagrange's Theorem

*more group theory*

$$(G : H) = \frac{|G|}{|H|}$$

---

## 120. Catalan Numbers

*A000108*

$$C_n = \sum_{k=1}^{n-1} C_k C_{n-k-1} = \frac{1}{n+1}\binom{2n}{n}$$

---

## 121. Ising Model Hamiltonian

*Mathematical model of ferromagnetism*

$$H(\sigma) = -\sum_{\langle i,j\rangle} J_{ij}\sigma_i\sigma_j - \mu\sum_j h_j\sigma_j$$

---

## 122. Borwein Integral

*The pattern famously breaks down after this integral.*

$$\int_0^\infty \frac{\sin(x)}{x}\cdot\frac{\sin\left(\frac{x}{3}\right)}{\frac{x}{3}}\ldots\cdot\frac{\sin\left(\frac{x}{13}\right)}{\frac{x}{13}}\,\mathrm{d}x = \frac{\pi}{2}$$

---

## 123. Wigner Semicircle Distribution

*Essentially just a semicircle scaled to be a probability distribution.*

$$f(x) = \begin{cases} \frac{2}{\pi R^2}\sqrt{R^2 - x^2} & -R \le x \le R \\ 0 & |x| > R \end{cases}$$

---

## 124. Parseval Gutzmer Formula

*Apply the Cauchy Integral Formula to derive*

$$f(z) = \sum_{k=0}^{\infty} a_k z^k \Rightarrow \frac{1}{2\pi}\int_0^{2\pi} \left|f\left(re^{i\theta}\right)\right|^2 \mathrm{d}\theta = \sum_{k=0}^{\infty} \left|a_k r^k\right|^2$$

---

## 125. Fubini's Theorem

*switching the order of integration ftw*

$$\int_X \left(\int_Y f(x,y)\,\mathrm{d}y\right)\mathrm{d}x = \int_Y \left(\int_X f(x,y)\,\mathrm{d}x\right)\mathrm{d}y = \int_{X\times Y} f(x,y)\,\mathrm{d}(x,y)$$

---

## 126. Coarea Formula

*A generalization of Fubini's theorem*

$$\int_{\Omega} g(x)|\nabla u(x)|\ \mathrm{d}x = \int_{\mathbb{R}} \left( \int_{u^{-1}(t)} g(x)\ \mathrm{d}H_{n-1}(x) \right)\ \mathrm{d}t$$

---

## 127. Equation of a Torus

*yum, donuts*

$$\left(\sqrt{x^2+y^2}-R\right)^2+z^2=r$$

---

## 128. Ampère-Maxwell law

*credit to Andrija*

$$\nabla \times \mathbf{B} = \mu_0 \left( \mathbf{J} + \varepsilon_0 \frac{\partial\ \mathbf{E}}{\partial t} \right)$$

---

## 129. Gauss's Flux Theorem (differential form)

*guess we're doing all of Maxwell's equations now huh*

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$

---

## 130. Gauss's law for Magnetism

*I'll need to fix this once we discover magnetic monopoles.*

$$\nabla \cdot \mathbf{B} = 0$$

---

## 131. Maxwell–Faraday equation

*induction*

$$\nabla \times \mathbf{E} = -\frac{\partial\ \mathbf{B}}{\partial t}$$

---

## 132. Eigenvalue Formula

*this yields the characteristic polynomial*

$$\det(\mathbf{A} - \lambda\ \mathbf{I}) = 0$$

---

## 133. Collatz Function

*The conjecture is that repeated applications of this function always hit 1.*

$$f(n) = \begin{cases} \frac{n}{2} & n \equiv 0 \pmod 2 \\ 3n+1 & n \equiv 1 \pmod 2 \end{cases}$$

---

## 134. Gamma Function

*A generalization of the factorial function*

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} \, \mathrm{d}x$$

---

## 135. Laplace Transform

*signals and systems baby*

$$\mathcal{L}\{f\}(s) = \int_0^\infty f(t) e^{-st} \, \mathrm{d}t$$

---

## 136. Taylor Series

*When $a = 0$, it's a Maclaurin series*

$$f(x) = \sum_{n=0}^{\infty} f^{(n)} \frac{a}{n!} (x-a)^n$$

---

## 137. Quaternion Multiplication Formula

*Hamilton famously carved this formula into the stone of a bridge when he came up with it.*

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{i}\,\mathbf{j}\,\mathbf{k} = -1$$

---

## 138. General Solution to First-Order Linear Differential Equations

*You can derive this with an integrating factor.*

$$y = e^{-\int P(x)\,\mathrm{d}x} \int Q(x) e^{\int P(x)\,\mathrm{d}x} \, \mathrm{d}x + C e^{-\int P(x)\,\mathrm{d}x}$$

---

## 139. Fibonacci Binomial Coefficients Identity

*Sum up the shallow diagonals of Pascal's triangle to make Fibonacci numbers*

$$F_{n+1} = \binom{n}{0} + \binom{n-1}{1} + \binom{n-2}{2} + \ldots + \binom{n - \lfloor \frac{n}{2} \rfloor}{\lfloor \frac{n}{2} \rfloor}$$

---

## 140. Bellman Optimality Equation

*Somehow connected to reinforcement learning! Credit to Constantine.*

$$V^{\pi *}(s) = \max_a \left\{ R(s,a) + \gamma \sum_{s'} P(s'|s,a) V^{\pi *}(s') \right\}$$

---

## 141. Definition of a Well-founded Relation

*R is well-founded iff every proper subset contains a minimal element with respect to R. Credit to Constantine.*

$$(\forall S \subseteq X)[S \neq \emptyset \Rightarrow (\exists m \in S)(\forall s \in S)\neg(sRm)]$$

---

## 142. Estimation Lemma

*Credit to Ben Napier.*

$$\left| \int_\gamma f(z) \, \mathrm{d}z \right| \leq L(\gamma) \sup_\gamma |f|$$

---

## 143. Chaitin's Constant

*The probability that a randomly constructed program will halt.*

$$\Omega_F = \sum_{p \in P_F} 2^{-|p|}$$

---

## 144. Cauchy's Differentiation Formula

*Credit to epm*

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_\gamma \frac{f(z)}{(z-a)^{n+1}} \, \mathrm{d}z$$

---

## 145. Defintion of the Quasi-Stationary Distribution

*Getting rid of absorbing states.*

$$\forall B \in \mathcal{B}(\mathcal{X}^a), \forall t \geq 0, P_\nu(Y_t \in B, T > t) = \nu(B) P_\nu(T > t)$$

---

## 146. Addition of Sound Levels in Decibels

*50dB + 50dB --> ~53dB!*

$$L_{ab} = 10 \log_{10}\left(10^{\frac{L_a}{10}} + 10^{\frac{L_b}{10}}\right)$$

---

## 147. Fast-Growing Hierarchy

*You wanna see some real speed?*

$$f_\alpha(n) = \begin{cases} n+1 & \alpha = 0 \\ f_\beta(n) & \alpha = \beta + 1 \\ f_{\alpha[n]}(n) & \text{else} \end{cases}$$

---

## 148. Feigenbaum-Cvitanović Functional Equation

*Damn, that's a mouthful.*

$$g(g(x)) = -\frac{1}{\alpha} g(\alpha x)$$

---

## 149. Dirac Equation

*Relativistic wave equation. Credit to Leon.*

$$i\hbar \gamma^\mu \partial_\mu \psi - mc\psi = 0$$

---

## 150. Feynman's Trick

*Essentially differentiating under the integral sign; the given problem is extremely difficult to solve otherwise. Credit to Aarsh Chotalia.*

$$\int_0^\pi \ln\left(1 - 2\alpha \cos x + \alpha^2\right) \, \mathrm{d}x = 2\pi \ln|\alpha|$$

---

## 151. Lorentz Factor

*Time and length change by a factor of gamma when objects move near the speed of light.*

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

---

## 152. Time Dilation

*Clocks moving at high speed will be observed to tick slower.*

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

---

## 153. Gauss's Flux Theorem (integral form)

*Use the divergence theorem to get to the differential form.*

$$\oiint_S \mathbf{E} \cdot \mathrm{d}\ \mathbf{A} = \frac{Q}{\varepsilon_0}$$

---

## 154. Doppler Effect

*beep beep beep*

$$\frac{f_o}{f_s} = \frac{\lambda_s}{\lambda_o} = \frac{v \pm v_o}{v \mp v_s}$$

---

## 155. Bernoulli's Equation

*I included just because it included this bonkers rho.alt thingy. What was wrong with rho??*

$$P_1 + \varrho g y_1 + \frac{1}{2} \varrho v_1^2 = P_2 + \varrho g y_2 + \frac{1}{2} \varrho v_2^2$$

---

## 156. Relation between $K_p$ and $K_c$

*Credit to Freddie Bullard.*

$$K_p = K_c (RT)^{\Delta n}$$

---

## 157. Van der Waals Equation

*Generalization of the Ideal Gas Law.*

$$\left(P + a\frac{n^2}{V^2}\right)(V - nb) = nRT$$

---

## 158. Maxwell-Boltzmann Distribution

*Don't have enough statistical mechanics formulas.*

$$f(v) = 4\pi v^2 \left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} e^{-m\frac{v^2}{2k_B T}}$$

---

## 159. Cayley-Hamilton Theorem

*Square matrices over commutative rings are annihilated by their own characteristic polynomial.*

$$p(\lambda) = \det(\lambda \mathbf{I}_n - \mathbf{A}) \Rightarrow p(\mathbf{A}) = 0$$

---

## 160. Chudnovsky's Formula for $\pi$

*This formula, based on a Ramanujan formula, was used to calculate pi to the tens of trillions of digits.*

$$\frac{1}{\pi} = 12 \sum_{k=0}^{\infty} \frac{(-1)^k (6k)!(545140134k + 13591409)}{(3k)!(k!)^3 (640320)^{3k+\frac{3}{2}}}$$

---

## 161. Residue Theorem

*Q: Why did the mathematician name her dog Cauchy? A: Because it left a residue at every pole.*

$$\frac{1}{2\pi i} \oint_\gamma f(z)\, \mathrm{d}z = \sum_{p \text{ pole}} \mathbf{I}(\gamma, p)\ \mathrm{Res}(f, p)$$

---

## 162. Center of Mass

*In a uniform gravitation field, this is the same as the center of gravity.*

$$\mathbf{R} = \frac{1}{M} \iiint_Q \rho(\mathbf{r})\ \mathbf{r}\ \mathrm{d}V$$

---

## 163. The Fundamental Group of the Circle

*It's isomorphic to the group of integers. Credit to fish.*

$$\pi_1(S^1) \cong \mathbb{Z}$$

---

## 164. Definition of the Operator Norm on a Finite Dimensional Banach Space.

*Credit to Richik Chakraborty.*

$$\left\{ \frac{\|T(x)\|'}{\|x\|} : x \neq 0, x \in X \right\} \equiv \{\|T(x)\|' : \|x\| = 1, x \in X\}$$

---

## 165. Green's Theorem

*Credit to Facejo.*

$$\oint_C (L \text{ d}x + M \text{ d}y) = \iint_D \left( \frac{\partial M}{\partial x} - \frac{\partial L}{\partial y} \right) \text{d}x \text{ d}y$$

---

## 166. Portfolio Variance

*Used to compute the covariance of a portfolio made up of $n$ different assets, if the single variances and covariances are known. Credit to Marco.*

$$\sigma_z^2 = \left( \sum_{i=1}^{n} w_i^2 \sigma_i^2 \right) + 2 \left( \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} w_i w_j \sigma_{i,j} \right)$$

---

## 167. Newton's Method

*Credit to https://github.com/lucasalavapena.*

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

---

## 168. Shannon Entropy

*Credit to https://github.com/lucasalavapena.*

$$H(X) = -\sum_{i=1}^{n} P(x_i) \log_2 P(x_i)$$

---

## 169. Pinsker's inequality

*It's possible I'm off by a factor of two here.*

$$\|\mu - \nu\|_{\mathrm{TV}} \leq \sqrt{2 D_{\mathrm{KL}}(\mu \parallel \nu)}$$

---

## 170. Sackur-Tetrode equation

*Entropy of monatomic ideal gas. Credit to Haydn Gwyn.*

$$\frac{S}{k_B N} = \ln\left[\frac{V}{N}\left(4\pi \frac{m}{3h^2}\frac{U}{N}\right)^{\frac{3}{2}}\right] + \frac{5}{2}$$

---

## 171. Condtional Entropy

*The amount of information needed to describe the outcome of a random variable given the outcome of another variable.*

$$H(Y|X) = -\sum_{x\in\mathcal{X}, y\in\mathcal{Y}} p(x,y)\log\left(\frac{p(x,y)}{p(x)}\right)$$

---

## 172. Force-Potential Relation

*Force is defined as the negative gradient of the potential energy function. Credit to Mayank Kumar.*

$$\mathbf{F} = -\frac{\partial U}{\partial x}\hat{\mathbf{i}} - \frac{\partial U}{\partial y}\hat{\mathbf{j}} - \frac{\partial U}{\partial z}\hat{\mathbf{k}} = -\vec{\nabla}(U)$$

---

## 173. Beta Function

*A special function that is closely related to the gamma function and to binomial coefficients. Credit to Salil Gokhale.*

$$B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\ \mathrm{d}t$$

---

## 174. Moist Adiabatic Lapse Rate

*The rate that the temperature falls with respect to altitude in a wet environment.*

$$\Gamma_{\mathrm{w}} = -\frac{\mathrm{d}T}{\mathrm{d}z} = g\frac{1 + H_{\mathrm{v}}\frac{r}{R_{\mathrm{sd}}T}}{c_{\mathrm{pd}} + H_{\mathrm{v}}^2\frac{r}{R_{\mathrm{sw}}T^2}}$$

## 175. Cardano's Formula

*Solution for a depressed cubic. Credit to TetanicRain7592.*

$$\sqrt[3]{-\frac{q}{2}+\sqrt{\frac{q^2}{4}+\frac{p^3}{27}}}+\sqrt[3]{-\frac{q}{2}-\sqrt{\frac{q^2}{4}+\frac{p^3}{27}}}$$

## 176. General Cubic Formula

*The deltas represents the cubic's discriminants. You must choose /any/ cube root and /any/ square root that doesn't result in C = 0. Credit to TetanicRain7592.*

$$C=\sqrt[3]{\frac{\Delta_1 \pm \sqrt{\Delta_1^2-4\Delta_0^3}}{2}}$$

## 177. Riemann Zeta Function

*This formula works when the real part of s is greater than 1. Other cases require analytic continuation.*

$$\zeta(s)=\frac{1}{\Gamma(s)}\int_0^\infty \frac{x^{s-1}}{e^x-1}\,\mathrm{d}x$$

## 178. Tangent Sum of Angles Formula

*Credit to TetanicRain7592.*

$$\tan(\alpha\pm\beta)=\frac{\tan(\alpha)\pm\tan(\beta)}{1\mp\tan(\alpha)\tan(\beta)}$$

## 179. Inner Product of Continuous Complex Valued Functions

*Credit to Zeus Hernández.*

$$\langle f,g\rangle=\int_0^{2\pi} f(t)\overline{g(t)}\,\mathrm{d}t$$

## 180. Definition of a Psuedorandom Generator

*Crypto means Cryptography!*

$$\left| \Pr_{x \leftarrow \{0,1\}^k} [\mathcal{A}(G(x)) = 1] - \Pr_{x \leftarrow \{0,1\}^{p(k)}} [\mathcal{A}(x) = 1] \right| < \mu(k)$$

---

## 181. Generalized Stokes' theorem

*One theorem to rule them all*

$$\int_{\partial M} \omega = \int_M \mathrm{d}\, \omega$$

---

## 182. Cartan's magic formula

*A cool little magic trick*

$$\mathcal{L}_X = \mathrm{d} \circ \iota_X + \iota_X \circ \mathrm{d}$$

---

## 183. Ridge Regression

*Estimate coefficients of a regression with L2 regularization*

$$\ell(\mathbf{w}) = \frac{1}{N} \|\mathbf{X}\ \mathbf{w} - \mathbf{y}\|_2^2 + \lambda \|\mathbf{w}\|_2^2$$

---

## 184. Evidence Lower Bound (ELBO)

*Lower bound on the log-likelihood of observed data*

$$L(\varphi, \theta; x) = \mathbb{E}_{z \sim q_\varphi(\cdot \mid x)} \left[ \ln \frac{p_\theta(x, z)}{q_\varphi(z \mid x)} \right]$$

---

## 185. Langevin Dynamics (Overdamped)

*Stochastic gradient flow with a deterministic potential and random noise*

$$\mathrm{d}x_t = -\nabla U(x_t)\ \mathrm{d}t + \sqrt{2\beta^{-1}}\ \mathrm{d}W_t$$

---